VALUE IN HEALTH 21 (2018) 1226–1233

Available online at www.sciencedirect.com

ScienceDirect

journal homepage: www.elsevier.com/locate/jval

Patient-Reported Outcomes

Accuracy of Linking VR-12 and PROMIS Global Health Scores in Clinical Practice

Brittany R. Lapin, PhD, MPH[1,2,*], Tyler G. Kinzy, MS[1,2], Nicolas R. Thompson, MS[1,2], Ajit Krishnaney, MD[3], Irene L. Katzan, MD, MS[1]

[1]Center for Outcomes Research and Evaluation, Cerebrovascular Center, Neurological Institute, Cleveland Clinic, Cleveland, OH, USA; [2]Quantitative Health Sciences, Cleveland Clinic, Cleveland, OH, USA; [3]Center for Spine Health, Neurological Institute, Cleveland Clinic, Cleveland, OH, USA

ABSTRACT

Objectives: To examine the accuracy of general health cross-walk tables in a clinical sample of patients with spine disorders. Published tables (Schalet BD, Rothrock NE, Hays RD, et al. Linking physical and mental health summary scores from the Veterans RAND 12-Item Health Survey (VR-12) to the PROMIS(®) Global Health Scale. J Gen Intern Med 2015;30:1524–30) link scores from the Veterans RAND 12-Item Health Survey (VR-12) to the 10-Item Patient-Reported Outcome Measurement Information System (PROMIS), a global health scale metric for both mental (GMH) and physical (GPH) summary scores. **Methods:** We assessed the accuracy of administered PROMIS and VR-12 scores with scores predicted by cross-walks in 4606 adult patients seen in a spine clinic from October 2015 to 2016. Accuracy of linking scores was evaluated using Pearson correlation, intraclass correlation coefficients, and mean and SD of score differences. Bland-Altman plots were used to graphically assess the levels of agreement. The consistency in scores' discrimination across levels of pain severity, depression, and other patient characteristics was assessed. Bootstrap methods estimated linking precision across varying sample sizes. **Results:** Actual and cross-walked PROMIS scores showed moderate correlation (ICC(3,1): GMH 0.73; GPH 0.81), with Bland-Altman plots suggesting smaller differences between scores in patients with lower and higher general health. Significant discrimination between patient subgroups was demonstrated reliably by both actual and estimated scores. Bootstrapped resamples indicated adequate precision for 200 patients (95% confidence interval for mean difference: GMH −1.38 to 0.60; GPH 0.39 to 1.93). **Conclusions:** VR-12 and PROMIS global health scores can be accurately linked within a sample of patients with spine disorders; nevertheless, bias is high and precision is low for linking on the patient level. Linked scores at the group level for more than 200 patients can be used in comparative effectiveness research and for comparing results across studies.

Keywords: cross-walk, patient-reported outcomes (PROs), PROMIS global health, score linking, spine, Veterans RAND 12-Item Health Survey (VR-12).

Introduction

Collection, tracking, measurement, and reporting of patient-reported outcomes (PROs) are increasingly important components of health care delivery. Information obtained directly from the patient can be used efficaciously to track burden of disease, assess outcomes of care, and compare effectiveness of different treatments. Also, importantly, PROs can be used as variables in risk adjustment models [1]. Global health measures are a type of PRO measure (PROM) that consists of an array of health items, each representing a different domain of health. Global health measures are designed to be concise, thus increasing the viability of routine administration. These are generic measures that are intended for use in patients with different conditions, allowing comparison across varying patient populations.

Two of the most commonly used global health measures are the Veteran RAND 12-Item Health Survey (VR-12) and the Patient-Reported Outcome Measurement Information System (PROMIS) Global Health Scale. The VR-12 is a freely available, self-administered health instrument primarily used to measure health-related quality of life, estimate disease burden, and evaluate disease-specific benchmarks with respect to directed populations [2]. It is typically used to assess health outcomes and administrative systems' performance in large health care systems, such as the Veterans Health Administration. The VR-12 is included in the Ambulatory Care Survey of Healthcare Experience of Patients [2] and the Health Outcomes Survey, which is an ongoing evaluation of the Centers for Medicare & Medicaid Services [3,4]. In addition, change in scores has been used as part of a risk-adjusted model within integrated health care networks [5,6].

* *Address correspondence to:* Brittany Lapin, Center for Outcomes Research and Evaluation, Cerebrovascular Center, Neurological Institute, Cleveland Clinic, 9500 Euclid Avenue, JJ3-603, Cleveland, OH 44195.
E-mail: lapinb@ccf.org

1098-3015$36.00 – see front matter Copyright © 2018, ISPOR-The Professional Society for Health Economics and Outcomes Research. Published by Elsevier Inc.
https://doi.org/10.1016/j.jval.2018.03.011

The PROMIS Global Health Scale, also known as the PROMIS-10, was developed more recently by the National Institutes of Health–funded PROMIS using the same psychometric techniques used in the development of other PROMIS tools [7,8]. The use of PROMIS-10 is increasing rapidly and has been considered as part of the National Health Interview Survey since 2010 and will be included in Healthy People 2020 [9]. The International Consortium for Health Outcomes Measurement has recommended PROMIS-10 for the measurement of outcomes in a range of conditions [10].

The use of different PROMs, such as the VR-12 and PROMIS-10, in research and clinical practice has posed a challenge when comparing outcomes across studies and populations. Differences in item content, response options, and scoring limit direct comparisons. One solution is to establish a linkage between scores of two different measures allowing calculation of scores across a common metric. This provides equivalent scores for different scales that measure the same construct. The National Cancer Institute funded the PROsetta Stone® project to link scores from many outcome instruments onto the PROMIS metric [11,12]. This initiative uses qualitative and quantitative methodology to link PROMIS global health scales with other global health measures [7,13]. To facilitate the implementation of PROMIS-10 and maintain continuity of previous research using the VR-12 measure, the PROsetta Stone project created cross-walks to link the VR-12 to PROMIS-10 [14]. Multiple linking methods were used to create the cross-walks on the basis of item-response theory (IRT) and equipercentile linking [15–17].

These cross-walk tables were developed using a general population sample and have not been validated in directed populations. Assessment of their performance in patients with differing characteristics and health statuses than the general population sample would provide important information on the utility of cross-walk tables for use in research and clinical care. Patients with spine disorders are a pertinent population for evaluating the accuracy of the linkage between the VR-12 and PROMIS-10. Lower back and neck pain was the leading cause of disability in most countries in 2015 [18]. Spinal surgeries are one of the most common surgical procedures in the United States [19]. Thus, the evaluation of PROs is especially important in patients with spinal disorders given the established health burden of this disease. Chronic low back pain was one of the six target conditions used in the Veterans Health Study's Medical Outcomes Study, on which the VR-12 was based, and has been used extensively as a measure of global health for patients with back pain [20]. With the increasing prevalence of PROMIS-10, there will be a growing need to link these two scores measuring similar constructs.

The aim of our study was to assess the accuracy of linking VR-12 scores to PROMIS-10 scores within the large clinical practice of patients with spine disorders at Cleveland Clinic (Cleveland, OH).

Methods

Study Design and Data

We performed a retrospective cohort study of patients seen in the Cleveland Clinic Spine Center who completed PROMs at the time of their visit. Through the Knowledge Program patient-entered data collection system, patients routinely complete electronic questionnaires before their visits to ambulatory clinics [21]. Patients are given the option to complete the questionnaires through a MyChart patient portal (Epic, Verona, WI) up to 5 days before their visit. If questionnaires are not completed beforehand, they are completed using tablet computers in the clinic lobby at the time of appointment check-in. The spine center collects patient-reported data using both generic quality-of-life and pain-specific instruments.

PROMIS-10 has been routinely completed in patients seen in the spine center since October 2015. PROMIS-10 produces two scores: global physical health (GPH) and global mental health (GMH) [8]. The GPH score comprises four items on physical health, physical functioning, pain intensity, and fatigue, whereas the GMH score includes four items on overall quality of life, mental health, satisfaction with social activities and relationships, and emotional problems. Two PROMIS global items (general health and social roles) are not used to calculate the physical and mental health summary scores. Summary scores are centered on the 2000 US Census with respect to age, sex, education, and race/ethnicity, and are transformed to a T-score metric with a mean of 50 ± 10 [22].

The VR-12 is a short-form version of the Veterans RAND 36-Item Health Survey, which was developed and modified from RAND-36 version 1.0 [2,23,24]. The VR-12 also produces physical and mental health summary scores on the basis of questions regarding physical functioning, role limitations due to physical or mental health problems, pain, energy, mental health, social functioning, and general health. As with PROMIS-10, physical and mental health summary scores are derived using an algorithm that is referenced to the 2000 to 2002 US Medical Expenditure Panel Survey population, and are standardized using a T-score metric with a mean of 50 ± 10 [2].

In addition to PROMIS-10 and VR-12, patients seen in the spine center completed a depression screen and a pain disability measurement. The Patient Health Questionnaire 9 (PHQ-9) is a nine-item depression screen frequently used in various patient populations. Its score ranges from 0 to 27, with higher scores indicating more depressive symptoms [25]. The Pain Disability Questionnaire (PDQ) was specifically developed for evaluating clinical outcomes in a population of patients with disabling musculoskeletal disorders, primarily involving the spine [26]. It yields a total functional disability score ranging from 0 (perfect function) to 150 (total disability).

Patient demographic data were obtained from the electronic health record. Approximate household income was estimated on the basis of 2010 Census data by zip code.

The study cohort consisted of adult (≥18 years) patients seen in the ambulatory spine center from October 11, 2015, to November 4, 2016, who completed all items on both PROMIS-10 and VR-12 at the same visit. If patients who met inclusion criteria were seen multiple times in this window, the first visit was used for analysis. This study was approved by the Cleveland Clinic Institutional Review Board (IRB no. 07-591).

Cross-Walk Data

Cross-walk tables to link VR-12 scores to PROMIS-10 scores were developed by Schalet et al. [14] using a general population sample collected by an Internet survey company including 2025 respondents (mean age 46 ± 18 years; 49% men). Before creating cross-walks, linking assumptions were verified to ensure that the two instruments measured the same construct, were highly correlated, and were unidimensional. Cross-walks were derived using multiple methods (IRT and equipercentile) for both raw summed scores and T scores. All published cross-walk tables were used in this study to directly link scores from the VR-12 to PROMIS-10. For reverse linking (PROMIS-10 to VR-12), linear interpolation was conducted to derive scores that were not included in the cross-walks using the equation $y = y_1 + (x - x_1) \times \frac{y_2 - y_1}{x_2 - x_1}$, where two known value pairs (x_1, y_1 and x_2, y_2) were used to determine the unknown y value for a given score x.

Statistical Analysis

Characteristics of patients with spine disorders are presented using descriptive statistics: frequency count with percentage for categorical variables and mean with SD or median with interquartile range for continuous variables.

Unidimensionality of the scales was assessed through OmegaH, an estimate of the proportion of total variance attributable to a general factor, and the explained common variance (ECV), the proportion of variance explained by one factor. Thresholds of 0.7 for OmegaH and 0.6 for ECV have been suggested as evidence for unidimensionality [27,28]. Internal consistency of scales was measured using Cronbach α.

The accuracy of linking was evaluated across all available methods provided by Schalet et al. [14]. Actual observed means with SDs were compared with the linked means with SDs, and mean differences between observed and linked scores were computed. Correlations were examined between actual and estimated scores using Pearson correlation coefficients and intraclass correlation coefficients (ICCs) as defined by Shrout and Fleiss [29] (ICC(3,1)). ICC was considered high at more than 0.90, moderate at more than 0.75, and low if less than 0.75 [30]. In addition, Bland-Altman plots graphically examined levels of agreement between the observed and linked scores [31].

We assessed the impact of varying sample size on linking estimate variability using bootstrap methods. An m-out-of-n resampling bootstrap procedure sampled subsets of m patients ranging from 10 to 500 by steps of 5. For each subset of size m, 10,000 replications were randomly sampled with replacement. An empirical 95% confidence interval (CI) of the mean difference between the actual and linked scores was computed. The half-width of the 95% CIs was reported as a measure of variability and precision.

Subgroup invariance was conducted by computing the standardized root mean square deviation, using the formula $\sqrt{\frac{\sum_{i=1}^{n}(\hat{y}_i-y_i)^2}{n}}$, between sex, age group (<65 years vs. >65 years), race, and marital status. In addition, means with SDs were compared between subgroups across observed and linked scores. Known-group validity by severity classification was assessed for level of depression and pain disability. Depression was classified according to the established PHQ-9 cutoff of 10+, indicating major depression [25]. Moderate/severe pain compared with no/mild pain was defined at a PDQ score of 71 or more, as determined by the literature [26]. Discrimination was assessed using Cohen d effect size, where values more than 0.2, 0.5, and 0.8 represent small, moderate, and large effects, respectively [32]. To compare Cohen d across severity for actual and linked scores, 95% CIs for the difference in effect size values were calculated [33].

Statistical analyses were conducted using SAS version 9.4 (SAS Institute Inc., Cary, NC) and R version 3.3.1 (R Foundation for Statistical Computing, Vienna, Austria).

Results

The demographic and clinical characteristics of the study sample of 4606 unique adult patients with spine disorders are presented in Table 1. The mean age was 57.5 ± 15.1 (range 18–102), with 34% of patients 65 years and older. The cohort was 47% males, 81% white, and 59% married. Six percent of patients had missing race or marital status, because oftentimes patients are uncomfortable providing this information. Almost 30% of patients indicated on the PHQ-9 that they were depressed, and 49% denoted moderate and/or severe pain on the PDQ.

In the participant data of Schalet et al. [14] that were used to create the cross-walks, the general population sample mean score on PROMIS-10 was 48.5 ± 10.0 for mental health summary scores and 48.3 ± 9.0 for physical health summary scores. Mean VR-12 T scores were 46.6 ± 11.1 and 45.2 ± 10.0 for mental and physical health summary scores, respectively. In comparison, our cohort of patients with spine disorders had similar mental scores for PROMIS-10 (46.2 ± 10) and VR-12 (47.4 ± 12) (Tables 2 and 3). Physical scores were lower for both PROMIS-10 (39.5 ± 8) and VR-12 (31.8 ± 11), which could be explained by the older age in our cohort as compared with the general population sample (57 ± 15 years vs. 46 ± 18 years) and because of the likely greater presence of spine-related symptoms and functional impairments in patients in this study cohort.

Cronbach α was high, ranging from 0.76 for PROMIS-10 GPH to 0.89 for VR-12 physical score, indicating high reliability of both measures. Factor strength estimates, OmegaH and ECV, were similar to those reported by Schalet et al. [14] and reached established thresholds, suggesting one general factor (data not shown).

Pearson correlation coefficients between actual PROMIS-10 and VR-12 scores were 0.73 for mental scores and 0.81 for physical scores. These coefficients also demonstrated a significant association between actual and estimated scores across all linking methods, with higher correlation seen within physical health summary scores compared with mental health summary scores (Tables 2 and 3). There was substantial variation in correlation between the linking methods, however. Actual and estimated PROMIS mental health summary scores showed low ICCs ranging from 0.727 for IRT preferred linking to 0.662 for the direct method of linking the VR-12 algorithm score to the PROMIS T score. PROMIS physical health summary scores showed moderate to low ICCs, with the highest for IRT preferred linking and the lowest for the direct method (ICC(3,1): 0.801 and 0.718, respectively). The mean difference, a measure of linking bias, between observed and estimated PROMIS mean scores using IRT linking was −0.40 ± 7.31 for the mental summary scores and

Table 1 – Characteristics of patients with spine disorders (n = 4606).

Characteristic	Total, N (%)
Sex	
Male	2145 (47)
Female	2461 (53)
Age (y), mean ± SD (range)	57.5 ± 15.1 (18–102)
Race	
White	3738 (81)
Black	496 (11)
Other	83 (2)
Unknown	289 (6)
Marital status	
Married	2702 (59)
Single	953 (21)
Other	668 (15)
Unknown	283 (6)
% below poverty*, median (q1, q3)	8.0 (6.1, 15.4)
Household income (×$10,000)*, median (q1, q3)	$53.1 (43.1, 65.8)
PHQ-9 score, median (q1, q3)	6 (2, 11)
Depressed (score 10+)	1081 (30)
PDQ score, mean ± SD	68.7 ± 35.6
Moderate/severe pain (score ≥71)	1183 (49)

PDQ, Pain Disability Questionnaire; PHQ-9, Patient Health Questionnaire 9.
* Estimated from zip code.

Table 2 – Actual and estimated PROMIS-10 scores based on published cross-walks using VR-12 scores (n = 4606).

PROMIS-10 scores	Global mental health summary scores				Global physical health summary scores			
	Mean ± SD	Pearson *r*	ICC	Mean difference (SD)	Mean ± SD	Pearson *r*	ICC	Mean difference (SD)
Actual PROMIS T score	46.2 ± 9.7	–	–	–	39.5 ± 8.0	–	–	–
Linked scores, methods								
IRT preferred linking	*46.6 ± 10.0*	*0.727**	*0.727*	*−0.40 (7.31)*	*38.4 ± 9.5*	*0.811**	*0.801*	*1.17 (5.54)*
Direct equipercentile								
No smoothing	46.4 ± 10.4	0.724*	0.723	−0.17 (7.49)	38.2 ± 9.2	0.807*	0.800	1.34 (5.44)
Less smoothing	46.3 ± 10.6	0.723*	0.721	−0.04 (7.58)	38.1 ± 9.8	0.806*	0.791	1.42 (5.78)
More smoothing	46.2 ± 10.6	0.722*	0.719	−0.03 (7.62)	38.1 ± 9.8	0.807*	0.792	1.46 (5.77)
Indirect equipercentile								
No smoothing	46.2 ± 10.1	0.727*	0.726	−0.01 (7.35)	38.3 ± 9.4	0.810*	0.799	1.21 (5.56)
Less smoothing	46.2 ± 10.2	0.726*	0.725	−0.03 (7.39)	38.1 ± 9.6	0.810*	0.798	1.42 (5.63)
More smoothing	46.2 ± 10.0	0.727*	0.727	0.04 (7.29)	38.1 ± 9.6	0.810*	0.797	1.46 (5.65)
Direct: VR-12 algorithm to PROMIS T score	49.5 ± 10.5	0.664*	0.662	−3.36 (8.31)	36.1 ± 9.1	0.724*	0.718	3.55 (6.42)

ICC, intraclass correlation; IRT, item-response theory; PROMIS, Patient-Reported Outcome Measurement Information System; VR-12, Veterans RAND 12-Item Health Survey.
Italicized text for IRT preferred linking indicates the optimal linking method.
* $P < 0.0001$.

1.17 ± 5.54 for the physical summary scores. The IRT method's SDs were among the lowest of all methods. Because means may be skewed in either direction, more emphasis was placed on the SDs between differences in actual and cross-walked scores.

The IRT linking method for estimating raw VR-12 scores from PROMIS-10 T scores resulted in higher Pearson correlation coefficients, ICCs, and smaller SDs when compared with the direct method (Table 3). This cross-walk resulted in a mean difference between observed and estimated VR-12 raw scores of 0.24 ± 4.33 for the mental component scores and −0.82 ± 3.55 for the physical component scores. Subsequent analyses describe results using the IRT linking method only.

Bland-Altman plots of observed versus estimated scores show that most of the mean differences were within 1 SD of the mean (Fig. 1). The mean differences close to 0 in all plots indicated no systematic variation in the differences. Linking bias was fairly high at the person level as signified by wide 95% limits of agreement between predicted and actual PROMIS scores, ranging from −14.72 to 13.92 for the mental scores and −9.70 to 12.04 for the physical scores, and VR-12 scores, ranging from −8.24 to 8.72 for the mental scores and −7.78 to 6.15 for the physical scores. For cross-walks in both directions (PROMIS scores linked to VR-12 scores, and vice versa), larger differences were seen for scores in the middle ranges. Accuracy appeared to be the greatest at the lower and higher ranges of the scales.

Bootstrapped resamples were used to calculate 95% CIs around the mean difference between actual and estimated scores across varying sample sizes. Figure 2 shows the half-width

Table 3 – Actual and estimated VR-12 scores based on linear interpolation from published cross-walks using PROMIS-10 scores (n = 4606).

VR-12 scores	Mental component score				Physical component score			
	Mean ± SD	Pearson *r*	ICC	Mean difference (SD)	Mean ± SD	Pearson *r*	ICC	Mean difference (SD)
Actual VR-12 raw score	22.6 ± 6.1	–	–	–	18.0 ± 6.0	–	–	–
Linked score, IRT preferred linking	22.4 ± 5.9	0.739*	0.739	0.24 (4.33)	18.8 ± 5.2	0.809*	0.801	−0.82 (3.55)
Actual VR-12 algorithm score	47.4 ± 12.4	–	–	–	31.8 ± 11.2	–	–	–
Linked score, direct PROMIS to VR-12 algorithm	43.5 ± 12.2	0.691*	0.691	3.87 (9.68)	36.2 ± 9.7	0.728*	0.721	−4.42 (7.82)

ICC, intraclass correlation; IRT, item-response theory; PROMIS, Patient-Reported Outcome Measurement Information System; VR-12, Veterans RAND 12-Item Health Survey.
Italicized text for IRT preferred linking indicates the optimal linking method.
* $P < 0.0001$.

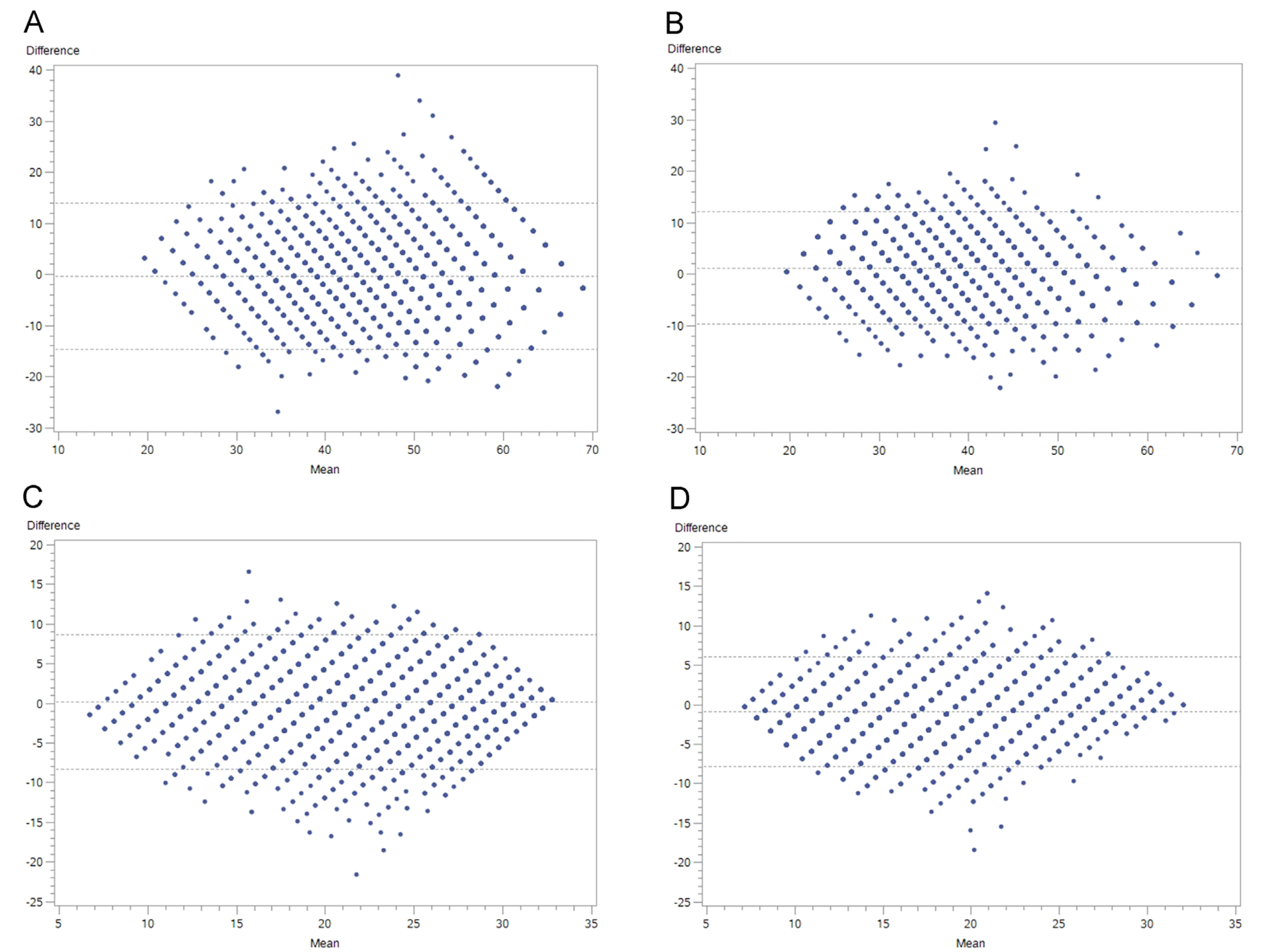

Fig. 1 – Bland-Altman plots of actual and estimated PROMIS global health T scores and VR-12 raw summed scores using the IRT preferred linking method (n = 4606). Bland-Altman plots of actual and cross-walked survey scores using IRT preferred linking, and linear interpolation for (A) PROMIS mental health T scores, (B) PROMIS physical health T scores, (C) VR-12 mental raw summed scores, and (D) VR-12 physical raw summed scores. The means of the actual and cross-walked scores are presented on the *x*-axis and the differences between the actual and cross-walked scores are presented on the *y*-axis. Horizontal dotted lines indicate the limits of agreement, or 95% CI, of the mean of differences. CI, confidence interval; IRT, item-response theory; PROMIS, Patient-Reported Outcome Measurement Information System; VR-12, Veterans RAND 12-Item Health Survey.

of the CIs. As sample size increased from 10 to 500, precision increased as seen by a decrease in variability around the mean difference. Half-widths of 1 and less were detected by n = 200 when linking to PROMIS-10 T scores, and by n = 75 when linking to VR-12 summed scores. Linking error was lower for physical scores than for mental scores.

As presented in Appendix Table 1 in Supplemental Materials found at https://doi.org/10.1016/j.jval.2018.03.011, actual and estimated PROMIS-10 mean scores were similar across patient sex, age group, race, and marital status. In addition, standard errors were comparable between actual versus linked scores across patient subgroups. Linkage invariance by demographic group has been suggested if less than 8% of the variance by subgroup is explained by instrument differences [34]. Standardized root mean square deviation values by sex, age group (<65 years vs. >65 years), race (white vs. black), and marital status (married vs. single) for observed and linked scores were all less than 6% (data not shown).

Both actual and estimated scores showed significant discrimination between patient health groups (Table 4). Effect sizes were large for differentiating between depressed and not depressed individuals, as well as between patients with moderate and/or severe pain versus no to mild pain. The linked scores had slightly higher effect sizes compared with the actual scores.

Reverse linking, PROMIS T scores to VR-12 raw summed scores, through linear interpolation demonstrated results similar to those in Tables 4 and Supplemental Table 1 (data not shown).

Discussion

Schalet et al. [14] report cross-walk tables to allow clinical researchers to convert VR-12 scores to PROMIS-10 global health scores and vice versa; these methods, however, were implemented in a general population. Our study demonstrates that the cross-walk tables maintain their accuracy in a clinical population of patients with spine disorders, across differing levels of health. The tables were created on the basis of multiple linking methods, with the IRT preferred linking method of summed VR-12 score to PROMIS T score proving the most

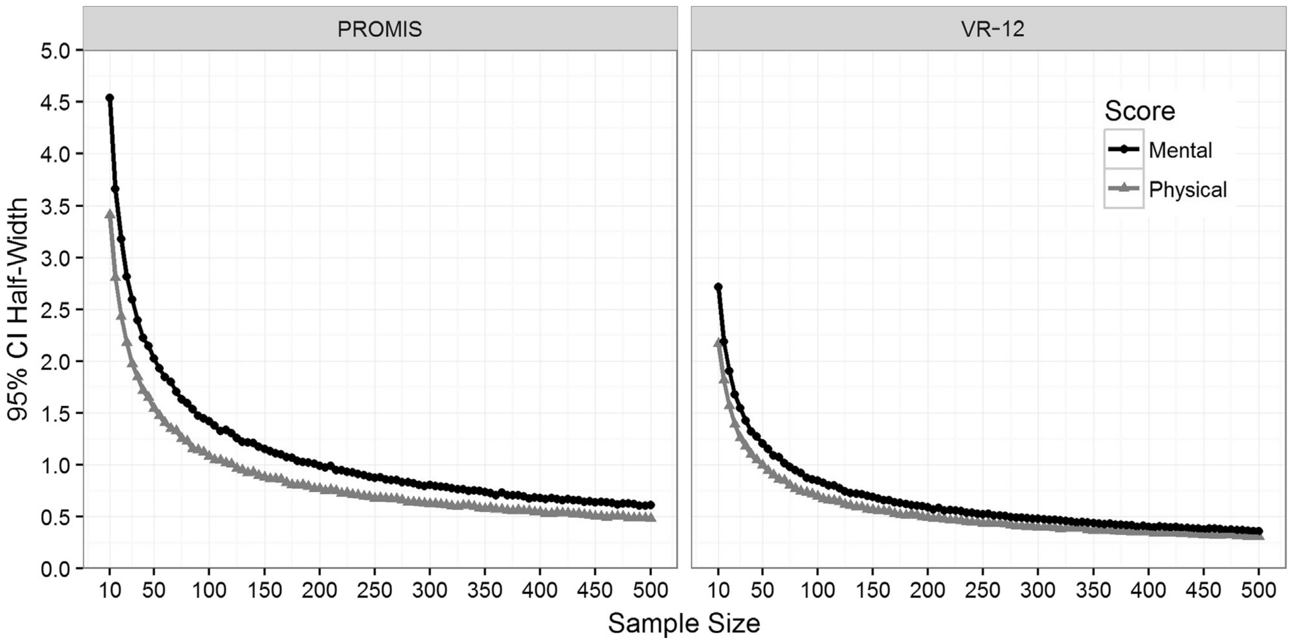

Fig. 2 – 95% CI half-widths between actual vs. estimated PROMIS global health and VR-12 scores across varying sample sizes. 95% CI half-widths are shown for the mean difference between actual and estimated scores across sample sizes ranging from 10 to 500 by steps of 5. The left panel presents actual compared with linked PROMIS mental and physical health T scores, and the right panel presents actual compared with linked VR-12 mental and physical raw summed scores. CI, confidence interval; PROMIS, Patient-Reported Outcome Measurement Information System; VR-12, Veterans RAND 12-Item Health Survey.

accurate [14]. Our study concurs and demonstrates excellent correlation between actual and linked scores using the IRT preferred linking method. These linking tables are easily implemented in clinical practice and do not require complex computer algorithms or specialized software. Our study agrees with the conclusion of Schalet et al. that for users who have item-level data, summed VR-12 items linked to PROMIS T scores have the highest correlations and the smallest SDs compared with linking based on algorithm-derived summary scores, especially for mental health summary scores.

Although the cross-walk tables were developed on the basis of data from the US general population, they function well in our specialized sample of patients with spine disorders with varying levels of health. The correlations between actual and linked scores were similar or higher in our study than in the general population sample. Schalet et al. found a correlation of 0.66 and SD of differences of 7.53 for the mental health summary score, and a correlation of 0.80 and SD of 5.90 for the physical health summary score. Despite having a cohort of older patients, our study of patients with spine disorders showed a higher correlation ($r = 0.73$) and a lower SD (7.31) for mental scores and similar correlation ($r = 0.80$) and SD (5.54) for physical scores, as compared with the general population they were based on. Our sample of patients is representative of patients with spine disorders seen at Cleveland Clinic Spine Center, a high-volume

Table 4 – Differences in actual vs. estimated PROMIS global health T scores by depression and pain disability status.

PROMIS-10 scores	Depression status[*]			Pain disability status[†]		
	Depressed, mean (SE)	Not depressed, mean (SE)	Effect size (95% CI)	Pain, mean (SE)	No pain, mean (SE)	Effect size (95% CI)
Total patients, N (%)	1081 (30)	2560 (70)	–	2179 (49)	2282 (51)	
PROMIS mental T score						
Actual	37.9 (0.24)	49.9 (0.16)	1.50 (1.43–1.58)	41.0 (0.19)	51.1 (0.17)	1.22 (1.15–1.28)
IRT linked	36.5 (0.21)	51.3 (0.15)	1.97 (1.89–2.05)	40.3 (0.18)	52.6 (0.16)	1.54 (1.47–1.61)
PROMIS physical T score						
Actual	33.1 (0.18)	42.4 (0.14)	1.35 (1.28–1.43)	34.1 (0.12)	44.8 (0.14)	1.79 (1.72–1.86)
IRT linked	30.4 (0.19)	42.1 (0.17)	1.49 (1.41–1.56)	31.8 (0.13)	44.6 (0.16)	1.84 (1.77–1.91)

CI, confidence interval; IRT, item-response theory; PDQ, Pain Disability Questionnaire; PHQ-9, Patient Health Questionnaire 9; PROMIS, Patient-Reported Outcome Measurement Information System; SE, standard error.

[*] Depression indicated by PHQ-9 scores: Depressed = PHQ-9 scores 10+; Not depressed = PHQ-9 scores <10.

[†] Pain disability status indicated by PDQ scores: Pain = PDQ scores 71+; No pain = PDQ scores <71.

center that treats a large and varied population. The wide geographic and demographic makeup of these patients increase the likelihood that generalizations can be made to a universal patient population with spine disorders [35,36].

Our study also showed consistent levels of convergent and discriminant validity between actual and linked mental and physical health summary scores. Actual and linked scores demonstrated subgroup invariance and ability to discriminate between depressed and pain-disabled patients.

Our research results have significant clinical utility. Clinical decision makers at medical centers using the VR-12 may wish to replace it with the newer PROMIS-10 instrument. Through the use of cross-walk tables, the historical data will not be lost. The linking will also facilitate comparative effectiveness research. The accurate aggregation of results from multiple outcome studies is important for use in systematic reviews and meta-analyses, which often comprise different instruments. We have demonstrated the accuracy of such pooling because aggregated results can be accurately converted from VR-12 to PROMIS metrics and vice versa. This will aid researchers to better understand the effects of treatment on global health.

Linking scores is a solution only for measures assessing the same unidimensional latent construct. Before undergoing the linking in this study, the content of both measures was evaluated by clinicians to ensure the measures were assessing the same underlying construct, global health-related quality of life, within patients with spinal disorders. Although the VR-12 and PROMIS-10 measure comparable constructs, linking global health surveys is challenging because of the breadth of the concept of global health and the observed differences in item content, wording, and format. Both our study and the study by Schalet et al. [14] yielded reliability indices and factor strength estimates that suggested that PROMIS-10 and VR-12 measure one single factor.

The scoring of these questionnaires is also complex as compared with the more typical approach based on raw sum scoring. PROMIS-10 is scored using IRT-based parameters as applied to each of the four items that contribute to the total summary score. Unlike the PROMIS-10 scales, the VR-12 components are weighted according to a factor analytic technique that constrains the components to be orthogonal, which may lead to variable total scores compared with simple sum scores [37]. The impact of this difference in scoring between PROMIS-10 and the VR-12 is apparent in the weaker accuracy for linking scores on the basis of the algorithmic approach. Nevertheless, the accuracy of using these cross-walk tables was confirmed in a population of patients with diverse spine disorders. Because potential error or bias is introduced by cross-walk procedures, it is recommended that cross-walks be used only for group-level comparisons and not to compare at the patient level [38]. As shown in our study, linking bias is substantially reduced if users of the cross-walk table convert scores of more than 200 patients and compare the mean of these converted scores.

Although our study clearly shows the accuracy and validity of the cross-walks in clinical practice at the group level, there are limitations to this research. Linear interpolation could not be conducted below the minimum observable cross-walk score. In our sample, a handful of patients had a minimum observed PROMIS-10 T score below the smallest listed cross-walk score, making linear interpolation impossible. As such, some lower PROMIS-10 physical health summary T scores could not be converted to VR-12 scores. Our Bland-Altman plots, however, indicated good agreement in the lower ranges. Another limitation is that linked PROMIS-10 to VR-12 scores, or vice versa, will have more error than scores obtained directly from the instrument because linking error is added to measurement error. Concordance between scores of any two instruments may be sensitive to population differences. The impact of these limitations will be mitigated by larger sample sizes. Because cross-walks are recommended for aggregate data on more than 200 patients, these limitations should not pose a problem in clinical use for comparing groups of patients. Nevertheless, linking should be a last resort for use in comparing studies or for conducting meta-analyses involving historical data. When determining a measure for use in clinical practice or research, content should be the foremost consideration.

Conclusions

VR-12 and PROMIS-10 global health scores can be accurately linked within a population of patients with spine disorders, although precision is low and bias is high for linking at the patient level. Our study recommends calculating linked scores directly between summed VR-12 scores and PROMIS T scores at the group level for more than 200 patients. The use of linked PROMIS-10 and VR-12 scores for individual patient management is not advised. Linked scores can be used in comparative effectiveness research and for comparing results across studies, increasing the generalizability of using the VR-12 and PROMIS-10 in clinical care and as PROMs in quality and research.

Source of financial support: No funding was received for this research.

Supplemental Materials

Supplementary data associated with this article can be found in the online version at https://doi.org/10.1016/j.jval.2018.03.011.

REFERENCES

[1] Ahmed S, Berzon RA, Revicki DA, et al. The use of patient-reported outcomes (PROs) within comparative effectiveness research: implications for clinical practice and health care policy. Med Care 2012;50:1060–70.

[2] Selim AJ, Rogers W, Fleishman JA, et al. Updated U.S. population standard for the Veterans RAND 12-Item Health Survey (VR-12). Qual Life Res 2009;18:43–52.

[3] Kazis LE, Selim AJ, Rogers W, et al. Monitoring outcomes for the Medicare Advantage Program: methods and application of the VR-12 for evaluation of plans. J Ambul Care Manage 2012;35:263–76.

[4] Hays RD, Sherbourne CD, Mazel RM. The RAND 36-Item Health Survey 1.0. Health Econ 1993;2:217–27.

[5] Selim AJ, Berlowitz D, Fincke G, et al. Use of risk-adjusted change in health status to assess the performance of integrated service networks in the Veterans Health Administration. Int J Qual Health Care 2006;18:43–50.

[6] Selim AJ, Kazis LE, Rogers W, et al. Risk-adjusted mortality as an indicator of outcomes: comparison of the Medicare Advantage Program with the Veterans' Health Administration. Med Care 2006;44:359–65.

[7] Cella D, Riley W, Stone A, et al. The Patient-Reported Outcomes Measurement Information System (PROMIS) developed and tested its first wave of adult self-reported health outcome item banks: 2005–2008. J Clin Epidemiol 2010;63:1179–94.

[8] Hays RD, Bjorner JB, Revicki DA, et al. Development of physical and mental health summary scores from the Patient-Reported Outcomes Measurement Information System (PROMIS) global items. Qual Life Res 2009;18:873–80.

[9] Black N, Burke L, Forrest CB, et al. Patient-reported outcomes: pathways to better health, better services, and better societies. Qual Life Res 2016;25:1103–12.

[10] The International Consortium for Health Outcomes Measurement Standard Sets. Available from: http://www.ichom.org/medical-conditions/. [Accessed April 18, 2018].

[11] Choi SW, Schalet B, Cook KF, Cella D. Establishing a common metric for depressive symptoms: linking the BDI-II, CES-D, and PHQ-9 to PROMIS depression. Psychol Assess 2014;26:513–27.

[12] Schalet BD, Cook KF, Choi SW, Cella D. Establishing a common metric for self-reported anxiety: linking the MASQ, PANAS, and GAD-7 to PROMIS Anxiety. J Anxiety Disord 2014;28:88–96.
[13] DeWalt DA, Rothrock N, Yount S, et al. Evaluation of item candidates: the PROMIS qualitative item review. Med Care 2007;45(Suppl. 1):S12–21.
[14] Schalet BD, Rothrock NE, Hays RD, et al. Linking physical and mental health summary scores from the Veterans RAND 12-Item Health Survey (VR-12) to the PROMIS(®) Global Health Scale. J Gen Intern Med 2015;30:1524–30.
[15] Kolen MJ, Brennan RL. Test Equating, Scaling, and Linking. New York, NY: Springer, 2004.
[16] Lord FM. The standard error of equipercentile equating. J Educ Behav Stat 1982;7:165–74.
[17] Reeve B, Fayers P. Assessing Quality of Life in Clinical Trials: Methods of Practice (2nd ed.). New York, NY: Oxford University Press, 2005.
[18] GBD 2015 Disease and Injury Incidence and Prevalence Collaborators. Global, regional, and national incidence, prevalence, and years lived with disability for 310 diseases and injuries, 1990–2015: a systematic analysis for the Global Burden of Disease Study 2015. Lancet 2016;388:1545–602.
[19] Fingar KR, Stocks C, Weiss AJ, Steiner CA. Most Frequent Operating Room Procedures Performed in U.S. Hospitals, 2003–2012. Healthcare Cost and Utilization Project Statistical Brief 186. Rockville, MD: Agency for Healthcare Research and Quality, 2014.
[20] Levy CE, Silverman E, Jia H, et al. Effects of physical therapy delivery via home video telerehabilitation on functional and health-related quality of life outcomes. J Rehabil Res Dev 2015;52:361–70.
[21] Katzan I, Speck M, Dopler C, et al. The Knowledge Program: an innovative, comprehensive electronic data capture system and warehouse. AMIA Annu Symp Proc 2011;2011:683–92.
[22] Liu H, Cella D, Gershon R, et al. Representativeness of the Patient-Reported Outcomes Measurement Information System internet panel. J Clin Epidemiol 2010;63:1169–78.
[23] Kazis LE, Lee A, Spiro A III, et al. Measurement comparisons of the Medical Outcomes Study and Veterans SF-36 Health Survey. Health Care Financ Rev 2004;25:43–58.
[24] Kazis LE, Miller DR, Clark JA, et al. Improving the response choices on the Veterans SF-36 Health Survey role functioning scales: results from the Veterans Health Study. J Ambul Care Manage 2004;27: 263–280.
[25] Kroenke K, Spitzer RL, Williams JB. The PHQ-9: validity of a brief depression severity measure. J Gen Intern Med 2001;16:606–13.
[26] Anagnostis C, Gatchel RJ, Mayer TG. The Pain Disability Questionnaire: a new psychometrically sound measure for chronic musculoskeletal disorders. Spine (Phila Pa 1976) 2004;29:2290–302.
[27] Reise SP. Invited paper: the rediscovery of bifactor measurement models. Multivariate Behav Res 2012;47:667–96.
[28] Reise SP, Bonifay WE, Haviland MG. Scoring and modeling psychological measures in the presence of multidimensionality. J Pers Assess 2013;95:129–40.
[29] Shrout PE, Fleiss JL. Intraclass correlations: uses in assessing rater reliability. Psychol Bull 1979;86:420–8.
[30] Wind H, Gouttebarge V, Kuijer PP, Frings-Dresen MH. Assessment of functional capacity of the musculoskeletal system in the context of work, daily living, and sport: a systematic review. J Occup Rehabil 2005;15:253–72.
[31] Bland JM, Altman DG. Statistical methods for assessing agreement between two methods of clinical measurement. Lancet 1986;1:307–10.
[32] Husted JA, Cook RJ, Farewell VT, Gladman DD. Methods for assessing responsiveness: a critical review and recommendations. J Clin Epidemiol 2000;53:459–68.
[33] Nakagawa S, Cuthill IC. Effect size, confidence interval and statistical significance: a practical guide for biologists. Biol Rev Camb Philos Soc 2007;82:591–605.
[34] Dorans NJ, Holland PW. Population invariance and the equatability of tests: basic theory and the linear case. J Educ Meas 2000;37 (4):281–306.
[35] Featherall J, Miller JA, Bennett EE, et al. Implementation of an infection prevention bundle to reduce surgical site infections and cost following spine surgery. JAMA Surg 2016;151:988–90.
[36] Lubelski D, Thompson NR, Agrawal B, et al. Prediction of quality of life improvements in patients with lumbar stenosis following use of membrane stabilizing agents. Clin Neurol Neurosurg 2015;139: 234–40.
[37] Farivar SS, Cunningham WE, Hays RD. Correlated physical and mental health summary scores for the SF-36 and SF-12 Health Survey, V.I. Health Qual Life Outcomes 2007;5:54.
[38] Dorans NJ, Pommerich M, Holland PW. Linking and Aligning Scores and Scales. New York, NY: Springer Science & Business Media, 2007.